

Governance / Constitution Debt

Status Audit

Date: 2026-06-14 **Scope:** Read-only verification of every §6.*-debt entry + CM-* paper-only/owed gate in root CLAUDE.md, cross-checked against the actual code/scripts/hooks at HEAD 357f867a. **Method:** Each row's "actual-status" is verified against a real artifact (file existence, hook line, verify-all wiring), not against the CLAUDE.md prose. Where CLAUDE.md prose disagrees with the artifact, the row is flagged STALE. **Anti-bluff (§11.4.6):** Every claim is cited to a file/line or "absent". No guessed status.

Debt-item status table

debt-id	requires	actual-status	evidence (file:line / absent)
§6.X-debt	Emulator PID inside podman/docker container managed by pkg/runtime; --runner=containerized for gate runs on every OS	PARTIAL	submodules/containers/pkgcontainerized.go exists with kvmDevicePath/kvmAvailabl (containerized.go:14-35); path on macOS still resolves RESOLVED_RUNNER="host-dim (scripts/run-challenge-matrix.sh:198,215). Linux x86_64+KVM path works; macOS host BLOCKED. Superseded by §6.AH-debt.
§6.Y-debt	Pre-push hook rejects code-touching commit after a distribute that did not bump versionCode	CLOSED	.githooks/pre-push:128-15 implements §6.Y bump-first against last-version-debug CLAUDE.md still lists it OWN STALE.
§6.Z-debt	firebase-distribute Phase-1 Gate 6	PARTIAL	Pre-push Check 7 enforces file presence on pointer ad

debt-id	requires	actual-status	evidence (file:line / absence)
	(evidence file: matching SHA + $\leq 24h$ + BUILD SUCCESSFUL) + pre-push check + tests/firebase/ hermetic test		(.github/hooks/pre-push:159-160) scripts/firebase-distribute.sh: NO runtime Gate 6 that validates timestamp/BUILD-SUCCESSFUL only a comment reference to distribute.sh:40-41). tests/firebase/ exists but tests version-modifier-pepper, not the §6.Z evidence (tests/firebase/).
§6.AA-debt	Default --debug-only; reject --release-only without companion debug evidence; split last-version-{debug,release}; tests/firebase/ hermetic test	CLOSED	firebase-distribute.sh:40 MODE="debug" default; :199 --release-only when last debug < current vc; split p :150,171. CLAUDE.md still PARTIAL → STALE.
§6.AB-debt	Detekt/go-vet rule for per-feature anti-bluff completeness (rendering/state-machine/gating)	OPEN	No Detekt rule; config/detekt contains only detekt.yml (detekt/ listing). The related check-challenge-discrimination exists+STRICT but covers C falsifiability markers, not completeness checklist.
§6.AC-debt	Detekt rule config/ detekt/lava-non-fatal-required.kts + go-vet lava-api-go/scripts/check-non-fatal-coverage.sh + pre-push + tests/non-fatal-coverage/	OPEN	config/detekt/lava-non-fatal-required.kts ABSENT; lava/scripts/check-non-fatal-coverage ABSENT; tests/non-fatal-coverage ABSENT. A bash scanner scans check-non-fatal-coverage (Kotlin-only, STRICT) but the Detekt custom rule + hermetic are not built.
§6.AD-debt (1) per-submodule pointer propagation	## INHERITED FROM block in all per-scope CLAUDE.md/ AGENTS.md/ CONSTITUTION.md	CLOSED	scripts/check-constitution 6.AD(4) enforces 54 in-scope (docs/helix-constitution-gates.md:56).
		CLOSED	scripts/check-commit-docs exists + wired in verify-all

debt-id	requires	actual-status	evidence (file:line / absence)
§6.AD-debt (2) CM-COMMIT-DOCS-EXISTS	verify referenced docs in commit messages resolve		verify-all-constitution-rules.sh:190). CLAUDE.md → STALE.
§6.AD-debt (2) CM-SCRIPT-DOCS-SYNC	script ↔ docs/scripts/*.md sync gate	CLOSED	scripts/check-script-docs-exists + verify-all :187; AN Check 9 (.github/hooks/pre-pu CLAUDE.md lists OWED →
§6.AD-debt (2) CM-SUBAGENT-DELEGATION-AUDIT	subagent delegation audit trail	CLOSED (scanner)	scripts/check-subagent-de audit.sh exists + verify-all doc + CLAUDE.md call it pa STALE.
§6.AD-debt (2) CM-COMMIT-DOCS-EXISTS/others equivalence-mapped	CM-ITEM-STATUS/TYPE-TRACKING, CM-FIXED-COLUMN-ALIGNMENT, operator-blocked gates	CLOSED-BY-EQUIVALENCE	§6.AD.3 Path B maps these ci-evidence/ ledgers (docs constitution-gates.md:14,16,19-22,36-3 Intentional; no scanner ow
§6.AD-debt (5) build-resource stats tracker	CM-BUILD-RESOURCE-STATS-TRACKER	CLOSED	scripts/build-stats-sampl scripts/build-stats-repor registry exist (docs/helix- constitution-gates.md:18) CLAUDE.md §6.AD-debt-5 : OWED → STALE.
§6.AE-debt	check-challenge-coverage.sh strict-mode flip + per-feature backfill	CLOSED (strict)	scripts/check-challenge-coverage.sh:39 STRICT="\$ {LAVA_CHALLENGE_COVERAGE_ (STRICT default); verify-all (:163). Gate-doc says "advi STALE. Runner host-gap is not §6.AE.
§6.AF-debt (2) chaos/stress §11.4.85 (LVA-7)	runnable chaos/stress beyond phase-1 lava-api-go	PARTIAL	scripts/run-chaos-stress (63 lines, scaffold); docs/ch DESIGN.md + EVIDENCE-phase NOT wired into verify-all (gate in scripts/verify-all constitution-rules.sh). Ph owed.
§6.AF-debt (LVA-6) codegraph own-	§11.4.79 — index own-org submodule SOURCE	CLOSED	.codegraph/config.json:3 records LVA-6 CLOSED 202 (1,842 submodule files ind submodule probe resolves

debt-id	requires	actual-status	evidence (file:line / absence)
org submodule inclusion			submodules/** exclude rem CLAUDE.md §6.AF lists OW STALE.
§6.AF-debt (4) per-clause CM-COVENANT-114-* -PROPAGATION gates	literal-anchor scan gates for §11.4.79–106	OPEN	Only CM-COVENANT-114-16-F row present (docs/helix-constitution-gates.md:14) clause propagation scanner new 79–106 anchors.
§6.AF-debt (5) §11.4.80 codegraph sync automation	auto git submodule init before index	OPEN	Precondition documented .codegraph/config.json:3 automation script that runs submodule init before code index.
§6.AG (no separate -debt)	Containers-driven emulators, no live ADB device	PARTIAL	Policy is in CLAUDE.md §6 enforced via run-challenge but the no-host-direct requirement superseded/tightened by § blocked by §6.AH-debt.
§6.AH-debt	Container/VM emulator path that BOOTS on macOS (no-KVM TCG mode + macOS /proc-reap fix in Containers submodule); no host-direct fallback	OPEN	containerized.go hard-req kvm for acceleration (containerized.go:19-21,3 no-KVM software-emulation run-challenge-matrix.sh:2 resolves macOS→host-direct recorded .lava-ci-evidence law-incidents/2026-06-03 boot-offline.json (present release canary + client gated BLOCKED on this macOS h
§6.AI-debt (1) §11.4.128 device recorder	background Genymotion/emulator recorder w/ deterministic layout	PARTIAL/ CLOSED?	scripts/record-device-se exists; tests/device-recor Whether it implements the deterministic YYYY-MM-DD/< hash>/... layout + gitignor codegraph-exclude was not verified — marked PARTIAL deeper check.
§6.AI-debt (2) §11.4.140 LAYER	UserPromptSubmit hook + generated slash commands	CLOSED (hook)	scripts/hooks/action-pre expand.sh exists + wired in settings.json:14-22

debt-id	requires	actual-status	evidence (file:line / absence)
2 action-prefix hook			(UserPromptSubmit); consti actions/registry.yaml pre Generated per-agent slash not verified. CLAUDE.md l → STALE for the hook port
§6.AI-debt (3) §11.4.141 thin-index restructure + token harness	de-dup 90K-token CLAUDE.md to index lines	OPEN	No evidence of a thin-index restructure; root CLAUDE.md full-body document.
§6.AI-debt (4) per-clause CM-COVENANT-114- {128..141}-PROPAGATION gates	literal-anchor gates for §11.4.128–141	OPEN	Not present in docs/helix- constitution-gates.md gat
CM-COVERAGE-LEDGER	§11.4.25 — strict-flip after baseline (Phase 7-debt)	CLOSED (strict)	scripts/check-coverage-1e STRICT=\$ {LAVA_COVERAGE_LEDGER_STR verify-all :156 invokes --s doc + CLAUDE.md mark it STRICT flip OWED" → STAI
CM-LVA-TICKETS-SYNC → CM-WORKABLE-ITEMS-SYNC	canonical workable-items binary; strict-flip	PARTIAL	scripts/check-workable-it exists + verify-all :202. Gat names the retired CM-LVA- SYNC/scripts/check-lva-ti (docs/helix-constitution- gates.md:25) → STALE; CLA §6.AF notes migration to c done. Upstream update/rec export pipeline OWED (no
CM-COVENANT-114-16-PROPAGATION + item-tracking paper-only gates	§11.4.15/16/19/21 trackers	CLOSED-BY-EQUIVALENCE	Intentional Path-B mapping helix-constitution- gates.md:14,19-24). Not tr

STALE-claim findings (CLAUDE.md/gate-doc says OPEN but artifact is CLOSED)

These are documentation drift, not real work. A master plan should NOT schedule work for them — only doc corrections:

- 1. **§6.Y-debt** — pre-push Check 6 is live (.github/hooks/pre-push:128). CLAUDE.md lists OWED.
- 2. **§6.AA-debt** — default flipped to --debug-only; release-only companion check live (firebase-distribute.sh:46,199). CLAUDE.md lists PARTIAL.
- 3. **§6.AD-debt items 2** — check-commit-docs-exists.sh, check-script-docs-sync.sh, check-subagent-delegation-audit.sh all exist + verify-all-wired + pre-push Check 9. Gate-doc marks several paper-only.
- 4. **§6.AD-debt item 5** — build-stats tracker shipped (build-stats-sample.sh/build-stats-report.sh). CLAUDE.md still lists OWED.
- 5. **§6.AE-debt** — check-challenge-coverage.sh is STRICT-default (:39). Gate-doc says advisory.
- 6. **CM- COVERAGE- LEDGER** — STRICT-default (check-coverage-ledger.sh:41, verify-all --strict). CLAUDE.md + gate-doc say advisory / Phase-7-debt OWED.
- 7. **LVA-6 (codegraph submodules)** — CLOSED per .codegraph/config.json:3. CLAUDE.md §6.AF lists OWED.
- 8. **§6.AI LAYER 2 action hook** — wired in .claude/settings.json:14. CLAUDE.md lists OWED.
- 9. **docs/helix-constitution-gates.md itself is stale** — "Last updated 2026-05-16"; predates §6.AF/AG/AH/AI (May 31 – Jun 9). Several rows describe pre-strict-flip state.

Truly-open debts (real work to schedule)

debt-id	one-line	effort	blocking
§6.AH-debt	macOS container/VM (TCG no-KVM) emulator path that actually boots; no host-direct fallback	L	YES
§6.X-debt	container-bound emulator gate path on macOS (folds into §6.AH)	L	YES

debt-id	one-line	effort	blocking
§6.AC-debt	Detekt lava-non-fatal-required.kts + Go-vet check-non-fatal-coverage.sh + tests/non-fatal-coverage/	M	no
§6.AB-debt	Detekt/go-vet per-feature anti-bluff completeness rule	M	no
§6.Z-debt (residual)	runtime Gate 6 inside firebase-distribute.sh (SHA/24h/BUILD-SUCCESSFUL) + hermetic test	M	partial
§6.AF-debt chaos (LVA-7)	chaos/stress §11.4.85 beyond phase-1; wire into verify-all	M	no
§6.AI-debt §11.4.141	thin-index restructure of CLAUDE.md + token-reduction harness	L	no
§6.AF/AI per-clause CM-COVENANT-**-PROPAGATION gates	literal-anchor scanners for §11.4.79–141	S	no
§6.AF-debt §11.4.80	auto git submodule init before codegraph index	S	no
§6.AI-debt §11.4.128	verify/complete deterministic device-recorder layout	S	no

Prioritized recommendations

1. **§6.AH-debt + §6.X-debt (TOP — both BLOCKING).** Single coordinated effort in the Containers submodule: add a no-KVM TCG software-emulation runner + the macOS /proc-reap fix so the container/VM emulator path boots on darwin/arm64. Until this lands, EVERY §6.AE matrix gate, §6.Z release canary, and tag-time attestation on the operator's macOS host is honestly BLOCKED — this is the single highest-leverage debt because it unblocks the entire release pipeline.
2. **§6.Z-debt residual (firebase Gate 6).** Add the runtime evidence gate inside firebase-distribute.sh (the pre-push Check 7 only fires

on the commit that advances the pointer, not at distribute time).

Cheaper than #1 and closes a real distribute-faulty-version vector.

3. **§6.AC-debt (non-fatal telemetry lint)**. Build the Detekt custom rule + Go-vet check + tests/non-fatal-coverage/. Currently only a Kotlin bash scanner exists; the Go side has zero enforcement. Pairs naturally with §6.AB-debt (same Detekt-rule-authoring effort).

Doc-hygiene action (not scheduled work, but should precede the plan): correct the 9 STALE claims above in CLAUDE.md + regenerate docs/helix-constitution-gates.md (stale since 2026-05-16) so the master plan is built on accurate status. Several debts are already done.

Open vs closed count: Of ~27 distinct debt/gate items examined: **10 truly-open** (2 BLOCKING, 1 partial-blocking, 7 non-blocking), **3 PARTIAL**, ****~14 CLOSED or closed-by-equivalence**** (9 of which are STALE-marked-open in CLAUDE.md/gate-doc).